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Studierichting: Informatica
Oefeningenreeks 2E nr. 10.2

$$28 \cdot 3^{x-1} - 3^{2x+1} = 1$$

$$\Leftrightarrow \frac{28 \cdot 3^x}{3} - 3 \cdot 3^{2x} - 1 = 0$$

$$3^x = t$$

$$\Leftrightarrow -3t^2 + \frac{28t}{3} - 1$$

$$D = \frac{784}{9} - 4 \cdot (-3) \cdot (-1) = \frac{676}{9}$$

$$t_{1,2} = \frac{\frac{-28}{3} \pm \frac{26}{3}}{-6} = \frac{1}{9} \text{ en } 3$$

$$3^x = \frac{1}{9} \Leftrightarrow x = -2 \text{ of } 3^x = 3 \Leftrightarrow x = 1$$

References